



main.c

 Share

Run

```
1 #include <stdio.h>
2 #include <math.h>
3
4 #define MAX 10000
5
6 int numSquares(int n) {
7     int squares[100];
8     int sqCount = 0;
9
10    for (int i = 1; i * i <= n; i++)
11        squares[sqCount++] = i * i;
12
13    int queue[MAX], front = 0, rear = 0;
14    int visited[MAX] = {0};
15
16    queue[rear++] = n;
17    visited[n] = 1;
18
19    int level = 0;
20
21    while (front < rear) {
22        int size = rear - front;
23        level++;
24
25        for (int i = 0; i < size; i++) {
26            int curr = queue[front++];
```

Output

Minimum perfect squares = 3

=== Code Execution Successful ===

```
main.c
24
25+   for (int i = 0; i < size; i++) {
26       int curr = queue[front++];
27
28+   for (int j = 0; j < sqCount; j++) {
29       int next = curr - squares[j];
30
31       if (next == 0)
32           return level;
33
34+   if (next > 0 && !visited[next]) {
35       visited[next] = 1;
36       queue[rear++] = next;
37   }
38   }
39 }
40
41
42 return level;
43 }
44
45+ int main() {
46     int n = 12;
47     printf("Minimum perfect squares = %d\n", numSquares(n));
48     return 0;
49 }
```

Output

Minimum perfect squares = 3

=== Code Execution Successful ===